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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

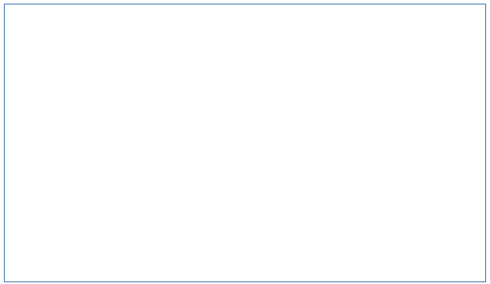
Lecture 23: Land use planning & Zoning

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Contents

- Land use planning
- Zoning
- Road patterns
- Planning models
- Network graph



Need for land use planning

- Land use planning is essential for any development project.
- Planning process can be used to prevent land-use conflicts and reduce exposure to pollutants.
- When it is performed thoroughly and accurately, the environmental, social, physical, and economic conditions will all benefit the local community

Example: Need for land use planning



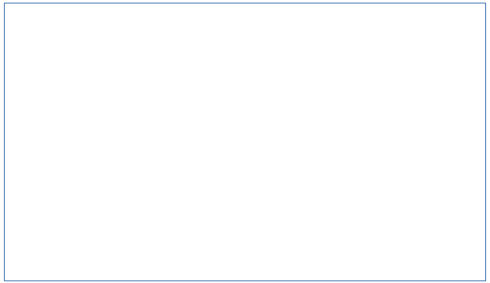
Dharavi at Mumbai is the world largest slum.

Do you think this can be changed using land use planning?

Yes, more compact, high rise with economic housing development can accommodate the population and have a potential to turn around the aesthetics of locality.

Definition of land use planning

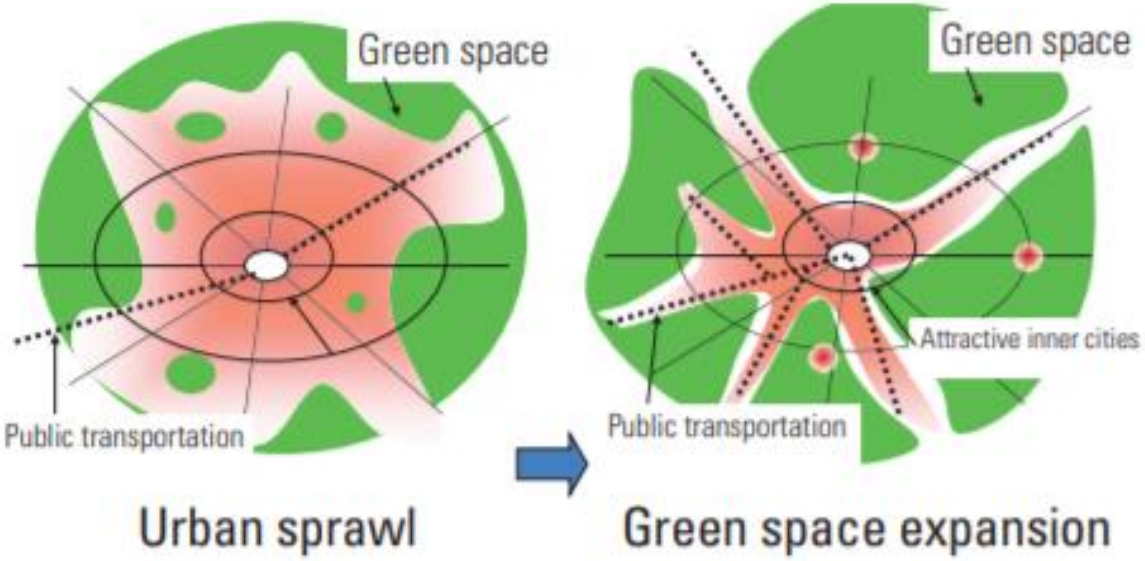
- Landowners, developers, municipalities, and other professionals collectively work to approve a land-based usage
- Promotes an appropriate social and environmental effect, while also employing resources efficiently
- Promote the orderly development of the landscape, preserve some open spaces, and provide consistency among developments



Objectives of land use planning

- Efficient utilization, acquisition and disposition of land
- Highest and best use of land
- Desirable patterns of land uses to prevent wasteful development and minimize the cost of public infrastructure and utilities and other social services
- Conservating the areas of ecological, aesthetic, historical and cultural significance

Example of efficient land use planning



Through land planning, we may utilize space efficiently and may use extra space for making locality greener and aesthetically pleasant.

Advantage of land use planning

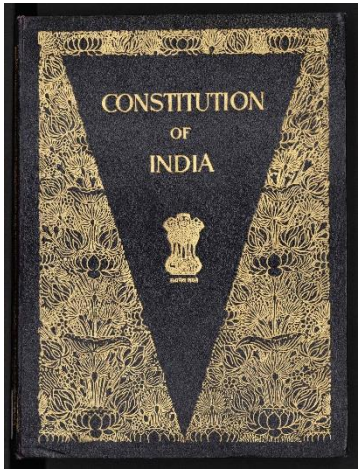
- Beneficial framework for the development
- Helps to anticipate the future of a particular area, implement an infrastructure that acts as risk mitigation
- Positively impacts the urban economy
- Put measures in place to combat climate change
- Protects lands from negative impacts of transportation
- Minimizes the public health and safety risks
- Prevents land use conflicts

Legal aspect of land use planning: Indian Constitution

Indian constitution refer land use planning at following sections:

- 74th constitutional amendment act:

Setting up and devolution of powers to Urban local bodies (ULBs) or city governments as the lowest unit of governance in cities and towns & planning rights.



- 243-ZE article: Metropolitan planning & rules for it.

- 243-ZD article: Committee for district planning & related rules.

Urban sprawling/organic growth of cities

- Rapid urbanization and migration towards cities is leading to expansion of urban areas.
- Expansion in cities boundaries are observed.
- This organic/unplanned expansion is called urban sprawling
- This leads to poor land use, poor lifestyle for residents and growth of slums near cities.



Image: Slum in sector 57, Gurugram (India)

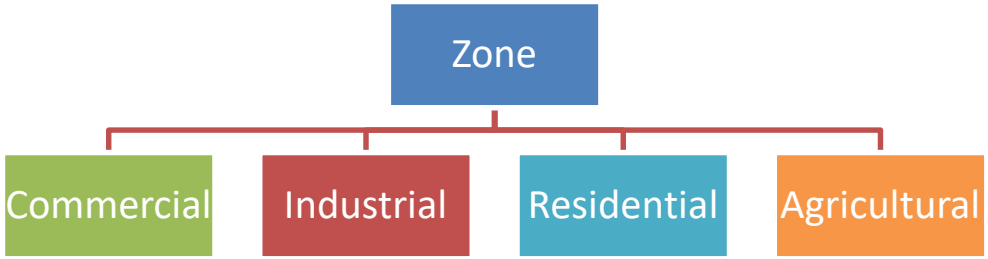
Image source: cityspidey.com

Zoning

- Zoning divides a city into multiple small parts.
- Zoning is the way the governments control the physical development of land and the kinds of uses to which each individual property may be put.
- Zoning is done for better governance and planning.
- Zoning is done based on socio-economic features.

Example: Noida, Greater Noida, Chandigarh, Gurugram are examples of zoned cities. Cities are divided into sectors.

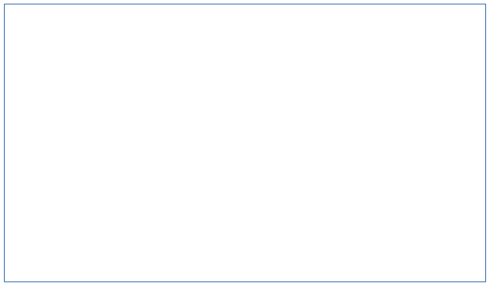
Zone classification



Based on socio-economic conditions, zones are classified mainly in these 4 categories (shown in figure)

Need of zoning: Example

- Frequency of metro/bus/shuttle services can be kept higher in morning time from residential to CBD (Central Business District) zone and vice-versa in evening.
- On off days (weekends), frequency towards markets zones can be kept high.
- In case of absence of clarified zone division, management of demand (frequency of public transport service) is difficult.

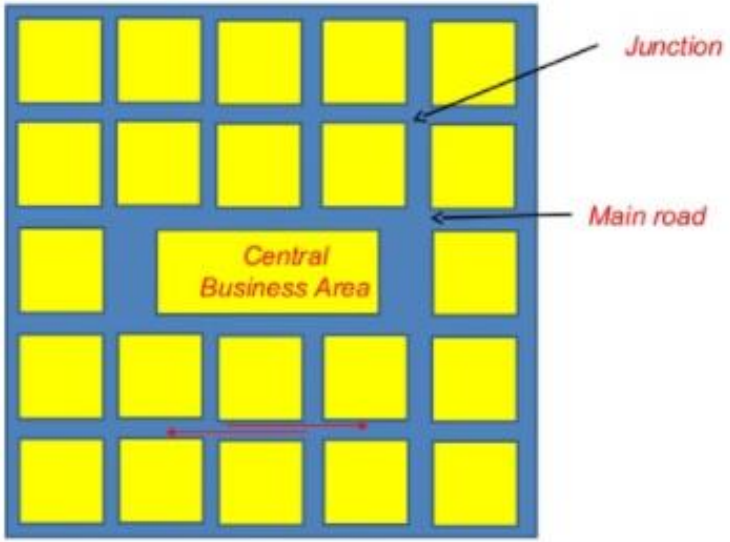


Road patterns based on land use

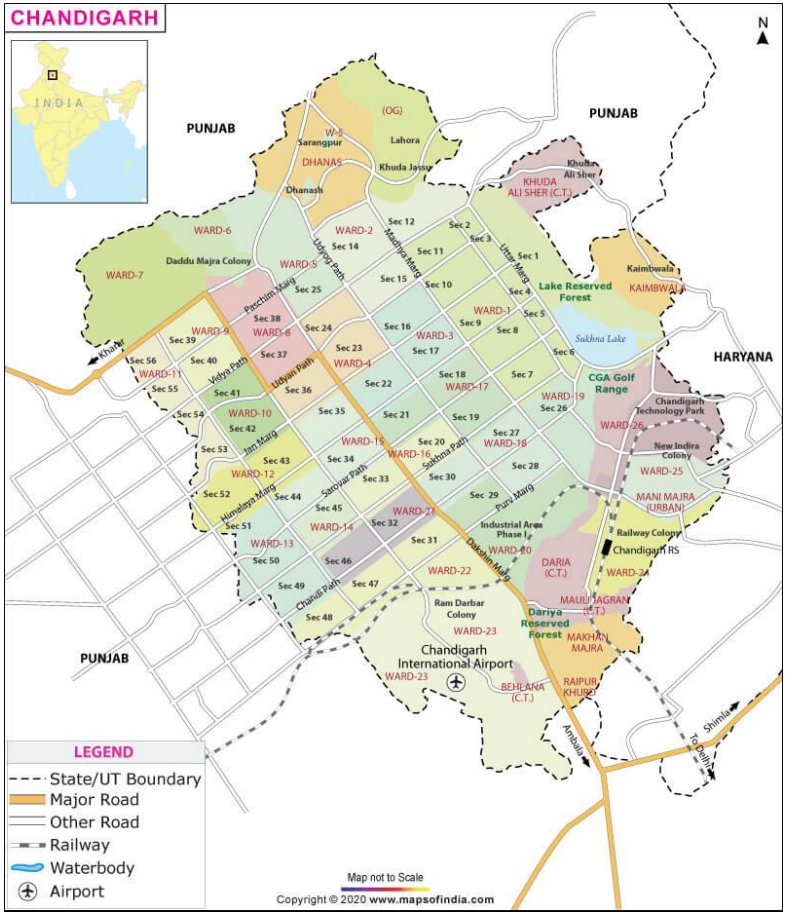
- Block pattern
- Star & block pattern
- Star & circular pattern
- Star & grid pattern
- Hexagon pattern

Block Pattern

- Whole area is divided into rectangular blocks
- Streets or branch roads intersect with each other at the right angle.
- The main roads always pass through the center, and it should be wide enough.
- Branch roads may be narrow as compared to main roads.
- The main roads should be provided with a direct approach to outside the city.



Example of block pattern: Chandigarh



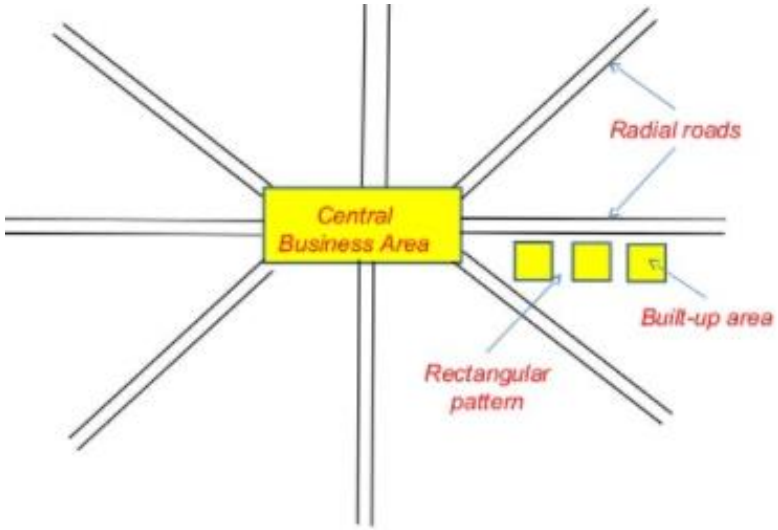
Road pattern is in rectangular form which is called blocked pattern in city of Chandigarh.

Image source: mapofindia.com

Star & block pattern

- Entire area is divided into a radial network of roads radiating from the center outwardly with block pattern network of roads in between the radial main streets.

- Reduces the level of congestion at the primary bottleneck location



Example of star & block pattern

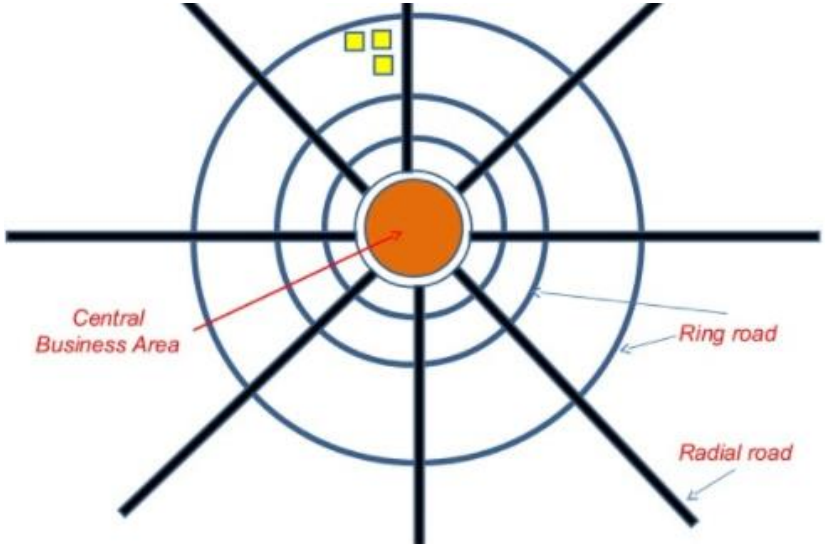


Image source: lonelyplanet.com

Image: *Arc de Triomphe (Paris, France)*

Star & circular pattern

- Main roads(radial roads) radiates from the central point and relate to concentric roads(ring roads) that are also radiating outwardly
- Reduces the possibility of rear-end crashes



Road network pattern in Delhi

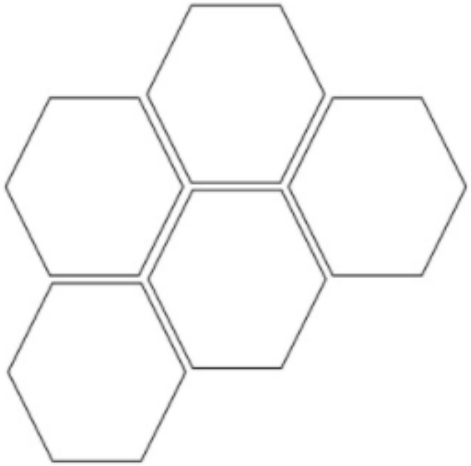


Image source: indiaonlinepages.com

Can you recognize pattern in central Delhi's road network (image)?

It is close to star & circular pattern

Hexagon Pattern



- Entire area is divided into hexagonal patterns.
- Three roads meet the built-up area boundary by the sides of the hexagons at every corner of the hexagon which can be further divided in suitable sizes.

Example of Hexagon Pattern

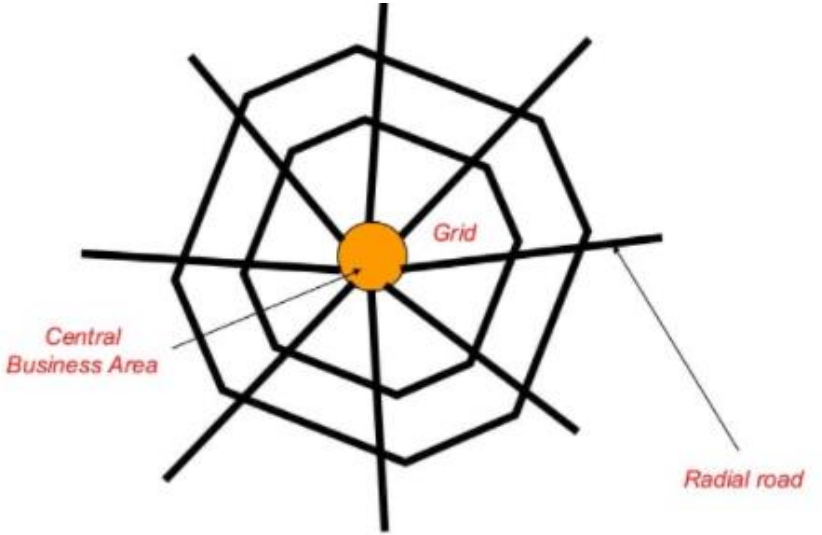


Image: *Grammichele, Italy* is a city with hexagon pattern land use and street network

Image source: wonderfulengineering.com

Star & grid pattern

- combination of Star and Grid Pattern.
- Radial network of road radiates from the center outwardly. Then, the main radial streets are interconnected by providing a grid pattern.

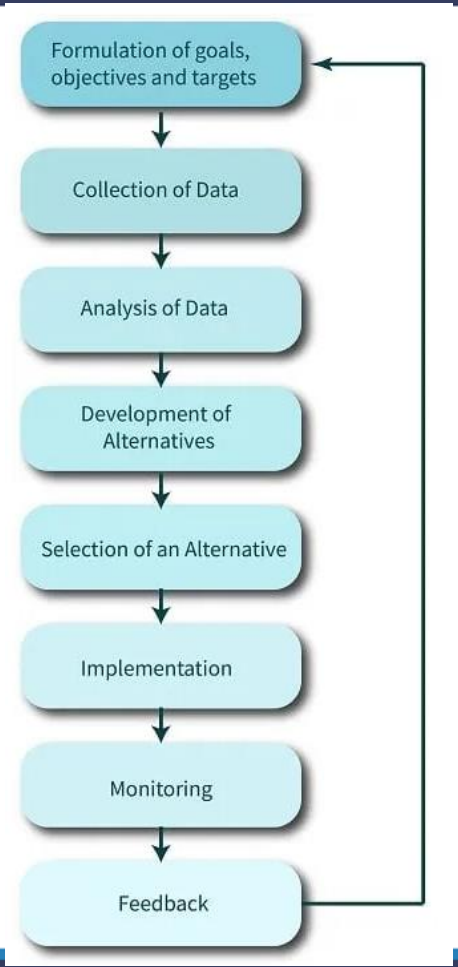


Planning models

- Rational planning model
- Advocacy planning concept
- Collaborative planning theory
- Political economic model in urban planning
- Critical minimum efforts theory

Rational planning model is most successful and commonly used method for planning

Rational Planning Model (RPM)



Considered as technocratic approach due to tedious data collection and dependency on technical expertise.

Do you think, this model is like EIA process/model?

Yes, it is
Conceptually both EIA process & RPM are decision making model for development

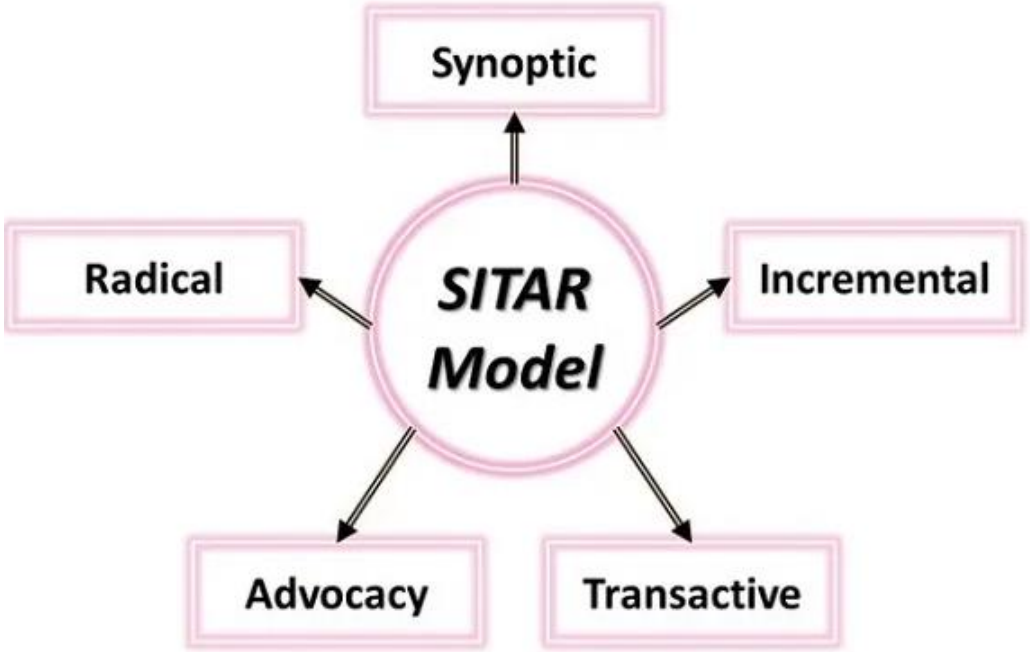
Source: planningtank.com

Advocacy/plural planning concept

Planning is done based on consultation with following:

- Representatives (mainly political)
- Interest groups (commerce of chamber, social groups, NGOs, think tanks etc.)
- Ad-hoc organization (temporary collaboration of people for any protest etc.)

SITAR planning concept: Land use planning model



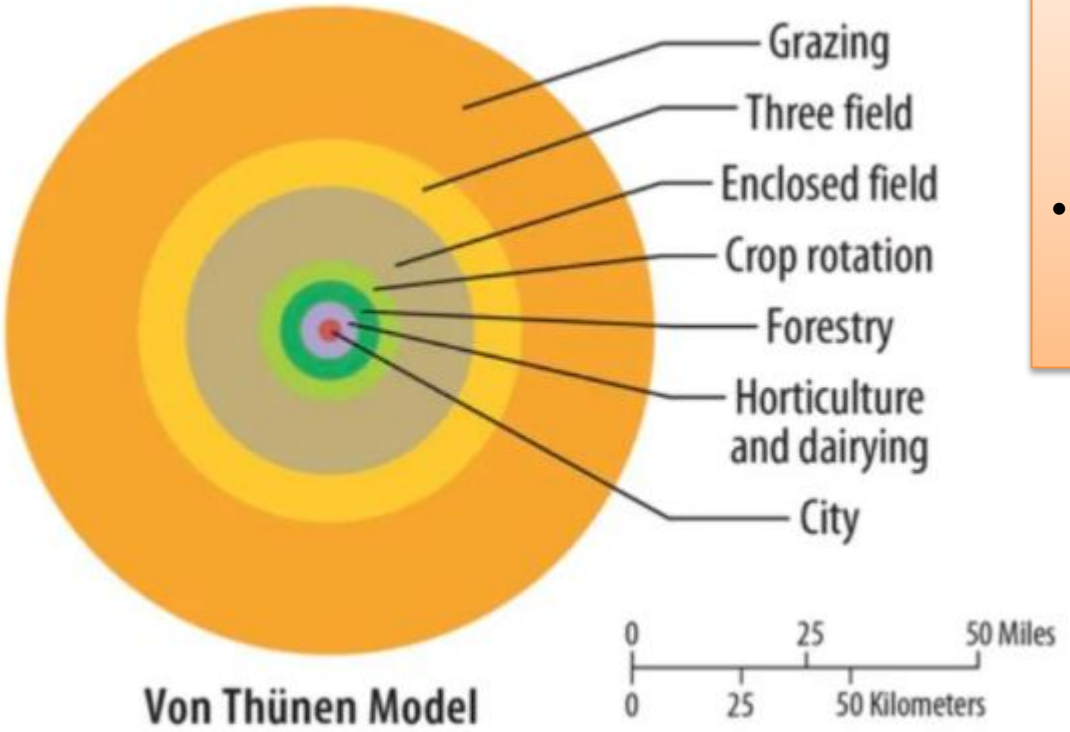
- Bunch of 5 conventions.
- Combination of other planning models.

Source: planningtank.com

SITAR planning concept: Terms

- Synoptic: Comprehensive in nature
- Incremental: Accommodating as per need
- Transactive: Replace the existing one with improved
- Advocacy: Views are taken up of different stakeholders
- Radical: Implementing new ideas

Von Thunen model for land use



Von Thünen Model

- This model was proposed for agricultural land use by Von Thunen in 1826.
- Land value or viability decreases with increase in distance from city.

Gentrification

- Gentrification is the rebuild of an area/locality into a more affluent and well-developed neighborhood.

- It is generally beneficial in terms of aesthetics & economy but may cause hardship for middle/low-income people.

Lisbon, Portugal –
It has become one of the most multicultural places & attracts more wealthy residents

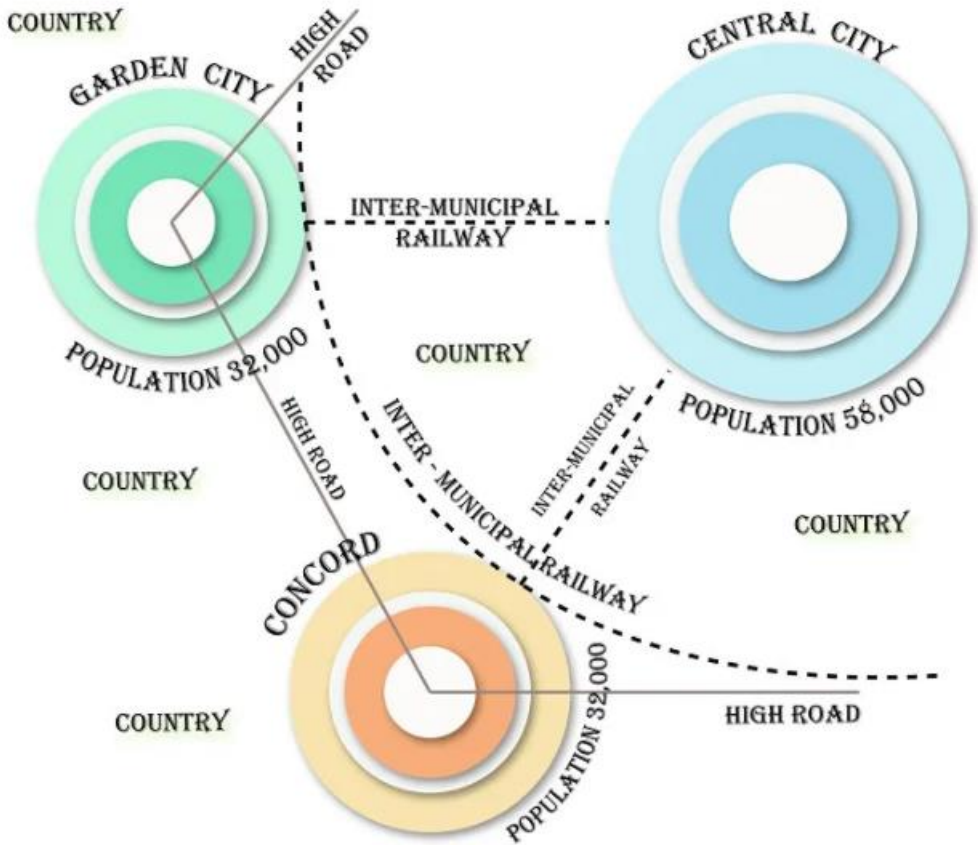
Lot of construction happening in the city. This makes it a gentrified city

Causing hardship among the middle-class population as it favors the upper middle class and affluent citizens

Garden city movement

- Utopian city suggested by Sir Ebenezer Howard in 1898.
- An environment friendly city (surrounded by green belt and appropriate space for gardens or open spaces)
- It is believed that it was the first modern day urban planning inspiration.
- Settlement or urban design that has been created using a holistic approach to planning and architecture

Design concept of garden city



Network Graph

Abstraction of network which is topologically identical to actual network, with all the properties of network embedded in it.

This embedding is done on the two elements that form the graph: nodes/vertices and links/edges

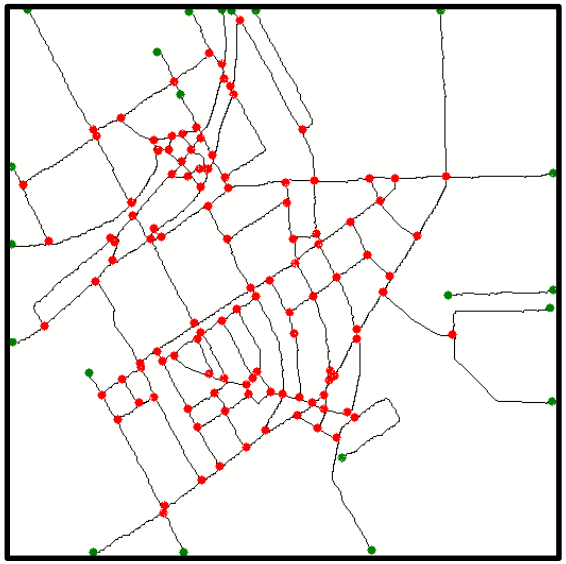


Image: Network graph of road network

Image source: Bhattcharya A., Indexing of satellite images using structural information

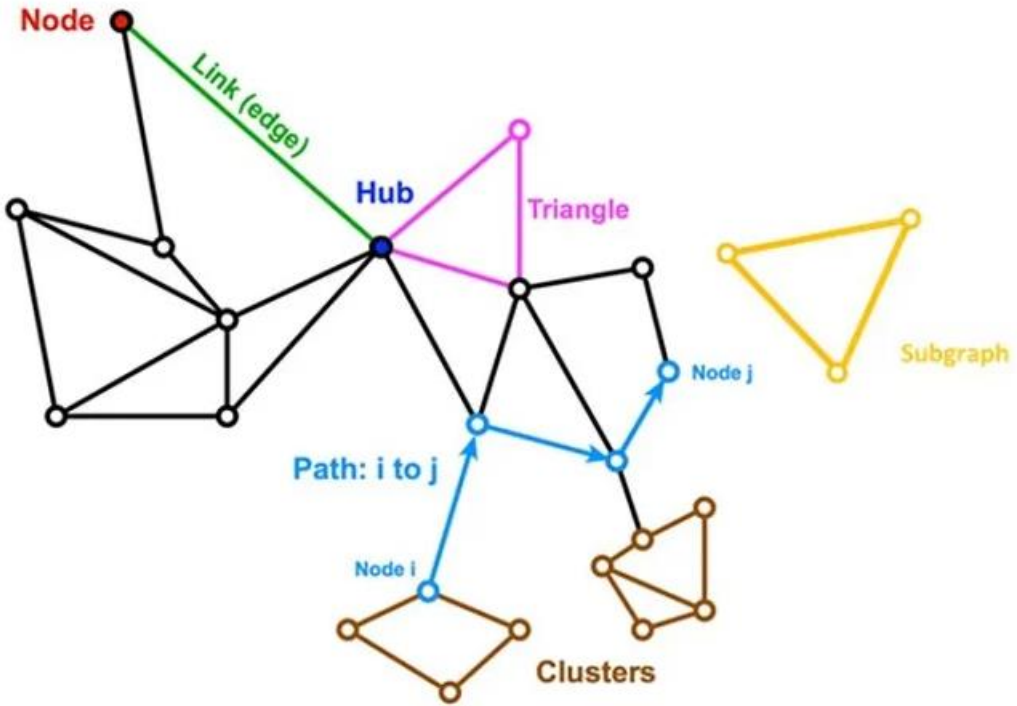
Node and link in network

A Node or vertex is a terminal point, or an intersection point on a graph.

It is the abstraction of a location such as a city, an administrative division, a road intersection, a dead end or a transport hub which connect to another network (stations, terminals, harbors and airports)

Link is the abstraction of a transport infrastructure supporting movements between nodes.

Network graph: Key words



Hub: Meeting points of 3 or more links.

Cluster: Closed loop of network

Sub graph: Independent part of network

Triangle: Closed loop with 3 nodes

Network Density

- Network density measures the territorial occupation of a transport.
- It is the length of the network per unit area.
- The higher the network Density, the more the network is developed.

$$NetworkDensity(ND) = \frac{Total\ link\ length\ (L)}{TotalArea(S)}$$

Eta Index (η)

- Eta index gives the average link length.
- Complex networks will tend to have a lower eta value.
- One can as well imagine it as how frequently a node will appear on average.

Walkable neighborhood, it's important for eta index to be lower allowing pedestrians more options for movement, but significantly reducing traffic speeds too

$$Eta\ Index(\eta) = \frac{Total\ link\ length\ (L)}{Tota\ lNo.\ of Links(e)}$$

Conclusion

- Land planning & transportation system are closely interlinked.
- Better integration of these benefits the users/residents.
- A proper planning model should be adopted for land use planning.

References

- Silberstein J., Mase C., 2013 “Land use planning for sustainable development”

- Brody S., 2013 “The Characteristics, Causes, and Consequences of Sprawling Development Patterns in the United States”; *Nature Education Knowledge* 4(5):2

Thank You